

TEACHER (large data)

STUDENT (small data)

Kaggle dataset
cardio_train.csv — 70,000 rows

UCI dataset
Cleveland heart disease — 303 rows

Teacher model
Strong XGBoost + calibration
Trained once, never sees UCI

Preprocess UCI
Fill missing values, SMOTE,
scale features

Soft labels (probabilities)
Teacher scores UCI patients 0→1

Feature selection
HHO / NGO / OOA algorithms
pick best feature subset

Student model
Lighter XGBoost on selected
features only

Dynamic fusion
Blend student + teacher prediction

Final prediction

